E2E Real Traffic Walkthrough — Architecture & Data Flow

System Architecture for Demo Walkthrough

```
flowchart TB
    subgraph Local Machine
        A[Browser - Frontend]
        B[Attack Scripts]
    end

    subgraph VPS - 187.124.45.161
        C[FastAPI Backend :8000]
        D[PostgreSQL]
        E[Redis]
        F[ML Worker]
        G[Capture Engine]
        H[PCAP Processor]
    end

    subgraph External Services
        I[OpenRouter - LLM]
        J[Threat Intel Feeds]
    end

    B -- SYN packets --> C
    B -- PCAP upload --> H
    H --> C
    C --> D
    C --> E
    E -- flows:live --> F
    F -- ml:scored --> C
    C -- async --> I
    I -- AI narrative --> C
    C -- IOC check --> J
    A -- REST API --> C
    A -- WebSocket --> C
```

Data Flow for Each Demo Step

```
sequenceDiagram
    participant U as User/Browser
    participant FE as Frontend
    participant BE as Backend API
    participant DB as PostgreSQL
    participant ML as ML Worker
    participant LLM as OpenRouter
```

```
    Note over U,LLM: Step 1: War Room
    U->>FE: Open /war-room
    FE->>BE: GET /flows/stats, /alerts/
    BE->>DB: Query data
    DB-->>BE: Results
    BE-->>FE: JSON response
    FE-->>U: Display metrics + map

    Note over U,LLM: Step 2: Attack
    U->>BE: nmap SYN packets
    BE->>DB: Store flows
    BE->>ML: Score via Redis
    ML-->>BE: anomaly_score
    BE->>DB: Create alert
    BE-->>FE: WebSocket new_alert
    FE-->>U: Alert appears in feed

    Note over U,LLM: Step 3: Alert Console
    U->>FE: Open /alerts
    FE->>BE: GET /alerts/id
    BE-->>FE: Alert + AI narrative
    FE-->>U: Display detail

    Note over U,LLM: Step 4: AI Analyst
    U->>FE: Send query
    FE->>BE: POST /llm/chat
    BE->>LLM: Stream request
    LLM-->>BE: Streaming tokens
    BE-->>FE: SSE stream
    FE-->>U: Typing effect
```

Component Status Assessment

Component	Expected	Current Status	Action Needed
Frontend	Running	Unknown	Verify localhost:3000 or Vercel
Backend API	Healthy	☑ Healthy	None
PostgreSQL	Connected	⚠ Pending	May need restart
Redis	Connected	☑ Healthy	None
ML Worker	Active	⚠ Idle	May need restart for live scoring
Capture Engine	Active	⚠ Idle	Expected - use PCAP upload
LLM Gateway	Available	Unknown	Test with chat query
Threat Intel	Synced	Unknown	Check IOC count

Risk Assessment

Risk	Probability	Impact	Mitigation
Database connection fails	Medium	High	Restart backend container
ML Worker not scoring	Medium	High	Use PCAP upload (pre-scored)
LLM API timeout	Low	Medium	Use cached narratives
Frontend build errors	Low	Medium	Use dev server instead
WebSocket disconnect	Low	Low	Page still shows data via REST

Recommended Approach

Given the current VPS state (ML Worker idle, Capture Engine idle), the **most reliable approach** is:

1. **Use PCAP upload for attack simulation** — bypasses capture engine, uses pre-processed heuristic scoring
2. **Verify existing alerts** — Tasks 1-2 already generated 590 alerts; use these for Steps 3-4
3. **Test live components only** — AI Analyst, Reports, Intel, Admin don't depend on live ML

This minimizes risk while still demonstrating the full E2E pipeline.